civilian team

لاب فيزياء

سارا العموش

أسئلة سنوات على آخر خمس تجارب :-

① The most accurate experiment to measure a resistor is :-
a. Ohms law exp. b. wheatstone bridge exp. c. RC-circuit exp.

الجواب هو "b"

② To convert the Galvanometer to ammeter we connect :-
a. Large resistor in series b. Large resistor in parallel
c. small resistor in parallel d. small resistor in series

الجواب هو "c"

③ If the potential difference across some resistor is doubled, then :
a. The resistance must be doubled b. The current must be doubled
c. The current must be halved d. The resistance must be halved

الجواب هو " b "

④ In charging capacitor. To increase the time of charging we must increase the value of :
a. The capacitance and/or resistance b. The capacitance (C)
c. The electromotive force of the power supply d. The resistance (R)

الجواب هو "b"

سارة العموش.

⑤ The most accurate experiment to measure a Large resistor is:-

a. Ohms Law exp. b. potentiometer exp. c. RC-circuit exp

الجواب هو "c"

⑥ The main aim of Potentiometer experiment is:-

a. test an old batteries b. design a voltage divider

c. increase the efficiency of the instruments d. measure the resistance with high accuracy

الجواب هو "d"

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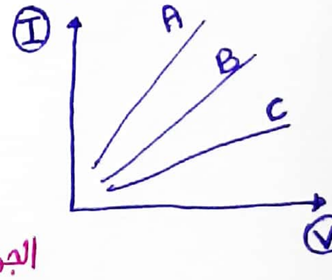
⑦ In ohm's Law exp. a student obtained the result shown in the figure for three trails: single resistor. two resistor in series and two resistors in Parallel, which of the following is true:-

a. A ⇒ series , B ⇒ single , C ⇒ Parallel

b. C ⇒ Series, B ⇒ Single , A ⇒ Parallel

c. A ⇒ series, B ⇒ Parallel, C ⇒ single

d. A ⇒ single, B ⇒ series, C ⇒ Parallel



الجواب هو "b"

⑧ To convert a Galvanometer of internal resistance $\boxed{R_G = 120\,\Omega}$ into voltmeter A large resistance of $\boxed{\overset{R_x}{20\,K\Omega}}$ connected in series with the Galvanometer. what is the maximum voltage in (V) can be measured by the constructed voltmeter. assuming that the maximum current sustained by the Galvanometer is equal $\boxed{\overset{I_{G,max}}{0.4\,mA}}$?

A. 5 V b. 20 V c. 8 V d. 12 V

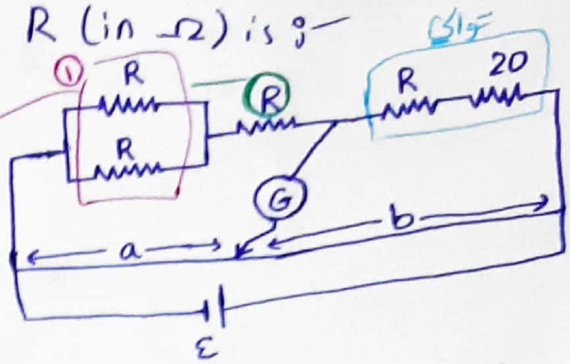
$$V_{max} = I_{G,max}(R_G + R_x) \Rightarrow V_{max} = 0.4\times10^{-3}(120 + 20\times10^{3})$$

$$V_{max} = 8\ V$$

الجواب هو "c"

(9) In the circut shown, if the galvanometer reads zero at $a = 12$ cm and $b = 25$ cm. The value of R (in Ω) is :-

a. 9.4 b. 20 c. 10 d. 40



$\frac{1}{R} : \frac{1}{R} = \frac{R}{2}$ بالتوازي

بالتوالي

$\frac{R}{2} + R = \frac{3R}{2}$

$$\frac{\frac{3R}{2}}{\boxed{R+20}} = \frac{12}{25} \Rightarrow \frac{3R}{2} \times \frac{12R + 240}{25}$$

$$75R = 24R + 480 \Rightarrow \boxed{R = 9.4}$$
$$-24R \quad -24R$$

الجواب هو "A"

سارة العويس

(10) A total resistance of $3.0\,\Omega$ is to be produced by combining two resistor, R and $12\,\Omega$ resistor. what is the value of R and how is it to be connected to the $12\,\Omega$ resistor?

a. $4\,\Omega$, series b. $4\,\Omega$, Parallel c. $2.4\,\Omega$, Parallel d. $2.4\,\Omega$, series

$$\frac{1}{12} + \frac{1}{R} = \frac{1}{3} \Rightarrow \frac{12+R}{\cancel{12}_{4}R} \times \frac{1}{\cancel{3}} \Rightarrow 12 + R = 4R$$
$$-R \quad -R$$

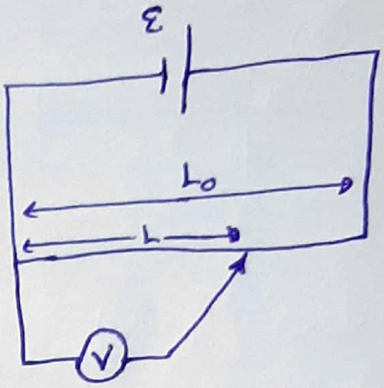
$$3R = 12 \Rightarrow \boxed{R = 4}$$

(R and $12\,\Omega$) connected in Parallel

الجواب هو "b"

(11) A Potentiometer was constructed as in figure shown, when the Pointer was at the mid of the given resistor, the voltmeter read 7 V. what is the voltage of the battery ε.

a. 7 V   b. 3.5 V   c. 6 V   d. 14 V



$\boxed{V = \varepsilon}$

الجواب هو "A"

(12) In discharging capacitor, if the voltage on it is halved after one minute. What is the value of the capacitance if it connect with 70 kΩ?

a. 0.9 mf   b. 830 F   c. 1.2 mf   d. 48.5 f

سارة العويش

$R = 70\,k\Omega = 70\times10^{+3}\,\Omega \;/\; t = 60\ sec$

$\boxed{V = \frac{V_o}{2}}$

$\frac{q(t)}{C} = \frac{Q_o}{C} e^{\frac{-t}{RC}}$ $V = V_{max}\, e^{\frac{-t}{RC}}$

$\frac{V_o}{2} = V_o\, e^{\frac{-60}{70\times10^3 C}} \Rightarrow \frac{1}{2} = e^{\frac{-6}{7\times10^3 C}} \Rightarrow Ln\, e^{\frac{6}{7\times10^3 C}} = Ln\, 2$

$Ln 2 = \frac{6}{7\times10^3 C} \Rightarrow C = \frac{6}{7\times10^3 \times \ln 2}$

$C = 1.2\times10^{-3} f \Rightarrow \boxed{1.2 mf}$

الجواب هو "C"

(13) To convert a Galvanometer of internal resistance $R_G = 120\Omega$ into ammeter a small resistance connected in Parallel with the Galvanometer using wire of resistivity equal $(300\,\Omega/km)$. If the maximum current can be measured by the constructed Ammeter is 3A, assuming the maximum current sustained by the Galvanometer is equal 0.2mA. what is the $\boxed{\text{length}}$ of the wire used to build up this ammeter?

a. 2.67cm b. 4 cm c. 3.5 cm d. 1.2 cm

$$R_s = \frac{I_{G\,max} R_G}{I_{max} - I_{G,max}} = \frac{(0.2\times10^{-3})\times(120)}{(3)-(0.2\times10^{-3})} = 0.008\,\Omega$$

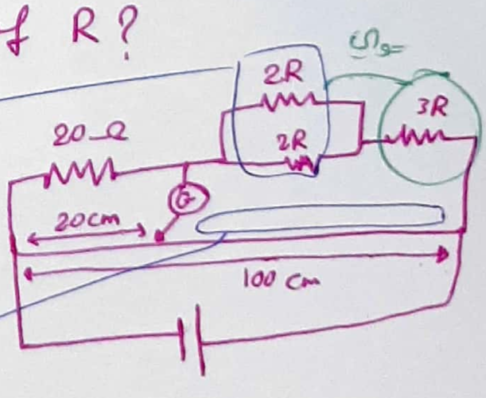
$$L = \frac{R_s}{\lambda}\times10^5\,cm = \frac{0.008}{300}\times10^5 = 2.67\ cm$$

سارة العموش

الجواب هو "a"

(14) In the "Wheatstone Bridge" exp. a student wanted to measure unknown resistance R. He found that I_G equal zero in the following state. what is the value of R?

توالي



$$\frac{1}{2R}+\frac{1}{2R}=\frac{2}{2R} \Rightarrow R$$ توازي

$$3R+R=4R$$

المطلوبة هاي =

$$b \leftarrow 80 = 20-100$$

$$\frac{4R}{20} = \frac{80}{80} \Rightarrow \boxed{R=20\Omega}$$

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